

**A FRAMEWORK FOR RETROSPECTIVE PROGRAM EVALUATION USING
FRAGMENTED EVIDENCE IN DATA-LIMITED INTERNATIONAL
DEVELOPMENT CONTEXTS**

A Literature Review Proposal
Presented

by

JEAN-FRANÇOIS JUTRAS

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INTRODUCTION

The growing interest in strengthening ex-post program evaluation has not fully resolved the methodological difficulties of its implementation. As Schimpf, Barbrook-Johnson and Castellani (2021) observe, “Despite two decades of increasing acceptance and use, implementing complexity-appropriate approaches to ex-post evaluation remains, in practice, a significant challenge” (p. 116). This highlights the continued relevance of this research area.

This literature review examines the theoretical and methodological foundations of ex-post program evaluation in international development, with particular attention to fragmented and incomplete information environments. It first presents the core principles of monitoring and evaluation systems and examines the role of ex-post evaluation in assessing results after implementation, especially when outcomes unfold over time and across multiple projects. It then discusses key constraints, including weak monitoring systems, missing or inconsistent data, limited continuity, organizational fragmentation, and the information paradox, where more data does not translate into usable knowledge.

The review then presents the main approaches to ex-post evaluation in practice, including criteria-based frameworks, quantitative, qualitative, mixed-method, and theory-based approaches. While these approaches are complementary, they assume that information is sufficiently structured to be combined across sources and levels of analysis. However, this condition is often absent, especially at program level. The review therefore identifies a gap in how fragmented project-level evidence can be integrated into coherent program-level evaluation. This gap motivates a more structured way of organizing and interpreting evaluative evidence under real-world data conditions.

1. THEORETICAL FOUNDATIONS OF EX-POST PROGRAM EVALUATION

1.1 Program Evaluation: Scope and Conceptual Foundations

Monitoring and evaluation (M&E) systems are widely used in international development because they help organizations track implementation progress and assess whether real change is occurring. Monitoring focuses on the continuous follow-up of activities and outputs during implementation, while evaluation focuses on assessing results, particularly outcomes and long-term impacts (Dijkzeul and Salomons, 2021; Glewwe and Todd, 2022; Samset and Volden, 2022). In applied settings, development programs are rarely single, isolated interventions. They are usually made up of multiple projects implemented across different locations and time periods. This creates the need to assess not only individual project performance but also how these components interact and what they produce collectively (Dijkzeul and Salomons, 2021).

Since the 1960s, program evaluation has evolved alongside the development of modern social science methods, becoming a standard tool in public administration and development practice (Langbein, 2014). It is mainly defined as a structured process used to determine whether an intervention has achieved its intended objectives and produced meaningful results. Evaluation also examines program design and implementation, and how these factors may have influenced observed outcomes (Langbein, 2014; Ponanan and Satirasetthavee, 2025). In this regard, evaluation is not only descriptive but also analytical, since it seeks to explain cause-and-effect relationships rather than only report indicators (Ponanan and Satirasetthavee, 2025).

Different evaluation approaches reflect different views on evidence, causality, and program complexity (Schiuma, 2015). Depending on the purpose and timing of the study, evaluations may use qualitative methods, quantitative methods, or a mix of both. They can also be conducted at different stages of a program: before implementation, during execution, or after completion, depending on the type of information needed (Khandker, B. Koolwal and Samad, 2009; Dijkzeul and Salomons, 2021; Glewwe and Todd, 2022).

1.2 The Role of Ex-post Evaluation

Ex-post evaluation refers to the assessment of programs or projects once their implementation has been completed (Khandker, B. Koolwal and Samad, 2009). It serves several purposes and has become increasingly important in international development and humanitarian contexts, where organizations are expected to demonstrate tangible results, learn from experience, and support decision-making with evidence rather than assumptions or intuition (Hallam and Bonino, 2013; Langbein, 2014; de Jong, Vignetti and Pancotti, 2019; Elgendy *et al.*, 2022). Unlike monitoring or evaluations conducted during implementation, ex-post evaluation focuses on what has actually been achieved once the intervention has ended and enough time has passed for outcomes to appear (Khandker, B. Koolwal and Samad, 2009). These evaluations are often carried out between one and five years after the start of a program or project (de Jong, Vignetti and Pancotti, 2019), although they may also be conducted at completion or shortly after (Linzalone and Schiuma, 2015).

Its relevance is also closely tied to accountability, transparency, and governance. By reviewing completed interventions, ex-post evaluation allows a critical assessment of

how resources were used and whether decisions produced meaningful results, while also strengthening institutional capacities (de Jong, Vignetti and Pancotti, 2019; Ponanan and Satirasetthavee, 2025). Its value depends on how far its findings are used in practice: when it is integrated into decision-making, it also contributes both to knowledge building and to more informed choices among stakeholders (Hallam and Bonino, 2013).

1.3 Methodological Requirements and Assumptions of Ex-post Program Evaluation

Ex-post program evaluation is commonly described as a rigorous process that depends on consistent data and effective monitoring systems (Khandker, B. Koolwal and Samad, 2009; Langbein, 2014; Glewwe and Todd, 2022). According to Glewwe and Todd (2022) evaluation begins with defining program objectives and identifying outcome variables. It also involves developing a theory of change, logic models, hypotheses, and performance indicators, while addressing issues such as validity, sampling, data collection tools, and data management systems. From this perspective, it is not only a retrospective activity but something that is embedded in program design and implementation from the beginning (Glewwe and Todd, 2022).

This is especially evident in quantitative approaches, where several methods are used to estimate program effects. Khandker, B. Koolwal and Samad (2009) identify techniques such as randomized evaluations, propensity score matching, double-difference estimators, instrumental variables, regression discontinuity designs, and structural modeling. Despite their differences, these methods share a common goal: identifying causal effects by building a credible counterfactual. This typically requires baseline data, comparison groups, or longitudinal datasets that make it possible to separate program

effects from other influencing factors (Khandker, B. Koolwal and Samad, 2009). This focus on structured evidence aligns with Langbein (2014), who emphasizes that evaluation relies on systematic comparison and credible empirical reasoning, both of which depend on well-organized information systems to support causal analysis.

The reliance on this type of information is not limited to quantitative approaches. Garbarino and Holland (2009) show that qualitative and mixed-method evaluations also require organized processes to capture and interpret results. Concepts such as empowerment, governance, or social capital still need to be operationalized through indicators and supported by coherent data collection strategies. While these approaches provide richer contextual understanding, they also depend on sufficiently structured information environments to produce robust and comparable findings (Garbarino and Holland, 2009).

Overall, a common assumption emerges across methodological traditions: ex-post evaluation typically relies on the existence of reliable, structured, and information-rich environments. Whether quantitative, qualitative, or mixed-method, evaluation frameworks depend on organized monitoring systems and available historical data that allow retrospective analysis and causal interpretation. However, these conditions are often absent in many international development contexts.

2 DATA CHALLENGES IN EX-POST EVALUATION

2.1 Institutional and Data Constraints

Despite its recognized value, the consistent application of ex-post program evaluation remains challenging in operational contexts. Existing research links these difficulties to a combination of institutional, temporal, and resource-related constraints that complicate retrospective assessment once interventions have ended (Rathinam *et al.*, 2021).

From an institutional perspective, development programs are often made up of multiple projects implemented across different periods and contexts. Ensuring coherent evaluation over time therefore depends on continuity in funding and organizational commitment. However, these conditions are rarely fully maintained after project completion, when resources are quickly redirected elsewhere (Späth, 2008; Hallam and Bonino, 2013; Castaño, Blanco and Martinez, 2019). These challenges are particularly strong in long-term development and infrastructure projects, where key outcomes may only become visible several years after implementation, creating a gap between planning, execution, and evaluation stages (Welde and Volden, 2018). Furthermore, organizational restructuring, staff turnover, and changes in information systems all contribute to disrupting data continuity and reducing the usability of evaluation findings (Welde and Volden, 2018; Dörrbecker, 2023).

These institutional constraints are closely connected to broader weaknesses in data availability and information management systems. Many studies highlight recurring issues such as missing baseline data, weak indicators, unclear objectives, fragmented

documentation, and poor archiving practices (Saqalli *et al.*, 2010; Nicolaisen and Driscoll, 2014; Welde and Volden, 2018; Dörrbecker, 2023; Li *et al.*, 2024). In several cases, relevant information on program design, outputs, and outcomes is only partially available or was never systematically collected in the first place (Langbein, 2014). This makes retrospective analysis difficult and can also lead to availability bias, where projects with more visible outcomes are more likely to be included in evaluation samples (Nicolaisen and Driscoll, 2014). Even qualitative and mixed-method approaches are not exempt from these constraints, as they still require sufficient data and resources to produce reliable findings (Garbarino and Holland, 2009; Castaño, Blanco and Martinez, 2019; Nguyen, 2024).

2.2 The Information Paradox in Development Contexts

At the same time, international development is marked by a clear paradox: while data production has increased significantly, important information gaps persist. Advances in monitoring technologies, donor reporting requirements, and results-based management systems have all contributed to growing volumes of data across the sector. Nevertheless, organizations often struggle to make effective use of this information, even when it is available (Obiorah, Ndubuisi and Aku Ugah, 2025).

This paradox is mainly explained by how data systems are designed and used rather than by a simple lack of data. In many cases, data collection is primarily driven by donor reporting needs rather than by organizational learning or decision-making priorities, which makes indicators often weakly connected to operational realities.

Monitoring systems also tend to operate in silos, leading to fragmented information that is not systematically integrated or fully used (Obiorah, Ndubuisi and Aku Ugah, 2025).

At the organizational level, limited analytical capacity and weak absorptive capacity further reduce the ability to transform data into actionable knowledge. This situation is reinforced by a focus on single-loop learning, where organizations mainly correct actions without questioning deeper assumptions or underlying systems.

Collectively, these factors point to the fact that the main challenge for ex-post evaluation is not only the absence of data, but also its fragmentation, uneven quality, limited integration, and weak institutional use (Obiorah, Ndubuisi and Aku Ugah, 2025).

3. REVIEW OF THE METHODOLOGICAL LANDSCAPE OF RETROSPECTIVE ASSESSMENT

3.1 Evaluating Organizational Development via Maturity Models

Although maturity models are not traditionally part of the ex-post program evaluation literature, they share several characteristics with evaluation frameworks. They commonly serve to assess organizational performance, identify strengths and weaknesses, and support learning and decision-making processes (Bititci *et al.*, 2015). Because of these similarities, maturity-oriented approaches are often used in organizational contexts as structured tools for reflecting on internal practices and development trajectories.

Lasrado, Vatrappu and Andersen (2015) describe three main ways maturity models are typically understood: as frameworks that map stages of organizational development, as best-practice-based guides, and as benchmarking tools that compare organizations against predefined maturity levels. Across these interpretations, a shared assumption is that organizations evolve in a relatively linear and staged way toward higher levels of capability. In this sense, maturity models are mainly developmental tools. They focus on capability building and structured improvement rather than on assessing specific interventions or reconstructing program outcomes over time. They also rely on structured and comparable performance information to track progression across maturity levels (Lasrado, Vatrappu and Andersen, 2015).

For this reason, maturity models are conceptually distinct from ex-post program evaluation frameworks, which focus on assessing the results and effects of specific interventions and require methods capable of supporting evaluative reasoning across time

and context (Dijkzeul and Salomons, 2021; Glewwe and Todd, 2022). This distinction may appear blurred in the institutional context of this research, where both approaches are sometimes mobilized for overlapping evaluative purposes. This section clarifies these conceptual differences to help better situate maturity-oriented approaches within the literature on retrospective assessment.

3.2 Structuring Heterogeneous Evidence via Multicriteria Decision Analysis

Multicriteria Decision Analysis (MCDA) methods are used to structure complex decision problems involving multiple and often conflicting criteria. They transform heterogeneous information into comparable and aggregable forms to support evaluation and decision-making (Žižović *et al.*, 2020; Amor *et al.*, 2023). Their primary function is therefore to organize information for comparison between alternatives, rather than to explain causal relationships or intervention effects (Alinezhad and Khalili, 2019; Amor *et al.*, 2023).

Methodologically, MCDA includes four main families of approaches: compensatory, distance-based, outranking, and weighting methods (Alinezhad and Khalili, 2019). In simple terms, these methods decompose performance into multiple criteria, assign weights to reflect their relative importance, and then aggregate them into outputs such as scores, rankings, or classifications. Some approaches allow trade-offs between criteria, while others rely on comparisons to reference points or pairwise evaluations. Despite these differences, they share the same underlying logic of decomposition, weighting, and aggregation. Their application depends on sufficiently detailed performance data and a clearly specified design to ensure consistent

operationalization of criteria and weights (Alinezhad and Khalili, 2019; Belahcène *et al.*, 2023a, 2023b, 2024).

In multi-project or program-level contexts, MCDA is often used to structure evaluation frameworks in which performance is defined through criteria such as cost, time, quality, or risk (Schiuma, 2015). Data are collected for each criterion, weighted according to their importance, and aggregated into composite outputs used to compare projects or interventions within a portfolio or program (Lauras, Marquès and Gourc, 2010). In sectors such as banking and real estate, similar approaches are used to combine multiple perspectives into utility functions reflecting different objectives, while composite indicators synthesize heterogeneous variables into a single evaluative measure (Ginevičius and Zubrecovas, 2009; Rebai, Azaiez and Saidane, 2016).

At an operational level, these applications can resemble ex-post evaluation in the way they structure information and aggregate performance across interventions. Still, MCDA remains primarily a tool for organizing and comparing information rather than for interpreting outcomes or explaining how they are produced (Kocaman and Asan, 2025). Ex-post program evaluation, by contrast, focuses on outcomes, effects, and their evolution over time across different contexts, and is grounded in a different analytical orientation.

3.3 Comparing Ex-post Evaluative Frameworks

This section examines existing ex-post evaluation frameworks at both project and program levels. This choice reflects the program–project logic highlighted in the literature, where programs are typically implemented through multiple projects (Dijkzeul

and Salomons, 2021), and responds to the observation that the boundary between project and program evaluation is not consistently defined across frameworks. Table 1 provides a comparative overview of key approaches across two dimensions: dependence on structured information and adaptability to fragmented evidence.

Table 1. Comparative Overview of Ex-post Evaluation Frameworks Under Varying Information Conditions

Study / Framework	Evaluation Level	Domain	Main Logic	Dependence on Structured Data	Adaptability to Fragmented Evidence	Main Limitations According to Authors
Crespo (2023)	Program	Public policies	Results-based OECD-DAC assessment	Moderate	Moderate-High	Cognitive biases and incomplete datasets
Dörrbecker (2023)	Project	Projects with fragmented data	Qualitative OECD-DAC expert judgment	Low	High	Evaluator subjectivity and variability
Samset and Volden (2022)	Multiple projects	Public investment: infrastructure and transport focus	Criteria-based OECD-DAC evaluation	Moderate	Moderate	Evaluator subjectivity and calibration issues
Späth (2008)	Project	Transdisciplinary research	Qualitative self-assessment logic	Low	High	Risk of bias due to self-assessment
(Bhagwat, Pathak and Venkataramani, 2011)	Program	Conservation policy	Mixed-methods, multidisciplinary, integrative framework	Low	High	Difficulty in establishing control groups and attributing causality
Mishra (2018)	Project	Infrastructure and transport	Social Cost-Benefit Analysis (SCBA)	Very High	Low	High dependency on structured data and subjective monetization of externalities
Welde and Volden (2018)	Multiple projects	Infrastructure and transport	Criteria-based OECD-DAC scoring	High	Moderate	Subjectivity in scoring and difficulty to measure long-term impacts
de Jong, Vignetti and Pancotti (2019)	Multiple projects	Infrastructure and transport	Hybrid CBA OECD-DAC scoring	High	Moderate-High	Subjectivity in scoring and methodological complexity
Ponanan and Satirasetthavee (2025)	Multiple projects	Infrastructure and transport	Hybrid CBA OECD-DAC scoring	Very High	Moderate	Balancing quantitative and qualitative scoring and difficulty to

						measure long-term impacts
Fadiga and Katjiuongua (2014)	Program	Animal health	Quantitative causal modelling	Very High	Low	Attribution challenges and data availability
Schimpf, Barbrook-Johnson and Castellani (2021)	Program	Policy and program evaluation	System simulation modelling (COMPLEX-IT platform)	Very High	Moderate	Model constraints and data uncertainty
Dershem, Komakhidze and Berianidze (2023)	Program	International development	Theory-based causal inference	Moderate-High	High	Context-specific findings and scalability
Vaz, Malaeb and Quinn (2019)	Program	Social policy and poverty alleviation	Composite index aggregation (Alkire-Foster)	High	Moderate	Complex and sensitive methodology

Note: The levels of dependence on structured data and adaptability to fragmented evidence were derived from the methodological characteristics, data requirements, and implementation conditions discussed in the reviewed studies.

The comparative review reveals three main groups of ex-post evaluation approaches. The first group is based on standardized criteria, particularly OECD-DAC-inspired frameworks, which rely on structured judgement, scoring systems, and relatively rapid assessments. As shown in Table 1, these approaches generally show low to moderate dependence on structured information and higher adaptability to incomplete or uneven evidence (Späth, 2008; Bhagwat, Pathak and Venkataramani, 2011; Samset and Volden, 2022; Crespo, 2023; Dörrbecker, 2023).

The second group includes more structured quantitative approaches, such as cost–benefit analysis, hybrid scoring systems, and integrated models. These methods improve comparability and measurement precision but require more consistently structured data, which increases their dependence on high-quality datasets and reduces flexibility when

information is fragmented (Fadiga and Katjiuongua, 2014; Mishra, 2018; Welde and Volden, 2018; de Jong, Vignetti and Pancotti, 2019; Vaz, Malaeb and Quinn, 2019; Ponanan and Satirasetthavee, 2025).

The third group consists of theory-driven and systems-based approaches, such as qualitative comparative analysis (QCA), process tracing, and simulation models. These methods are better suited to handling complexity and incomplete information, but their reliance on structured data varies, and they often face limitations in comparability, scalability, and cross-case aggregation due to the context-specific nature of their results (Schimpf, Barbrook-Johnson and Castellani, 2021; Dershem, Komakhidze and Berianidze, 2023).

Across these groups, Table 1 shows that the relationship between structured information and adaptability is not linear. Some frameworks prioritize standardization and comparability, which requires more formalized inputs, while others put the emphasis on flexibility and contextual interpretation under weaker information conditions. This reflects different ways of managing the trade-off between structure and interpretive depth rather than a simple gradient of data quality. It also shows that the distinction between project-level and program-level evaluation is not always clear in practice, as program evaluations often rely on project-level evidence without clearly specifying how it is aggregated across interventions and contexts.

Overall, the literature reveals a fragmented landscape of ex-post program evaluation rather than a single unified framework under varying information conditions. Approaches differ in how they structure and use information, balance standardization and flexibility, and translate project-level evidence into program-level conclusions. This

leaves the challenge of integrating heterogeneous evidence into a coherent program-level evaluation unresolved, particularly through a single approach (Schiuma, 2015).

3.4 Identifying the Methodological Gap

To summarize the preceding discussions, ex-post program evaluation plays a central role in international development as one of the main tools for assessing long-term results and overall program effectiveness after implementation (Dijkzeul and Salomons, 2021; Samset and Volden, 2022). It is particularly relevant in contexts where outcomes emerge over time across multiple projects, and several authors argue that it should be more systematically applied given its importance for understanding long-term and cross-project effects in complex program environments (Longhurst, 2013; de Jong, Vignetti and Pancotti, 2019; Dörrbecker, 2023). As highlighted in the previous sections, ex-post evaluation remains difficult to implement in practice, especially in settings characterized by weak continuity and uneven information systems (Nicolaisen and Driscoll, 2014; Dijkzeul and Salomons, 2021; Dörrbecker, 2023).

These difficulties are mainly driven by two recurring conditions. First, relevant information is often incomplete, as key elements such as baseline data, consistent indicators, or outcome measures are missing because they were not collected, standardized, or preserved over time. Second, even when information exists, it is rarely integrated, since evidence is dispersed across projects, reporting systems, and institutional layers, making it difficult to reconstruct a coherent program-level picture (Nicolaisen and Driscoll, 2014; Welde and Volden, 2018; Dijkzeul and Salomons, 2021; Dörrbecker, 2023). Building on this, monitoring systems are often designed for

accountability and reporting purposes rather than for evaluative synthesis, which reinforces fragmentation over time (Obiorah, Ndubuisi and Aku Ugah, 2025).

Existing evaluation approaches address these conditions in different ways. Experimental and quasi-experimental methods aim to identify causal effects through structured data and comparison designs, but they depend on stable systems and high-quality baseline or longitudinal data (Khandker, B. Koolwal and Samad, 2009; Langbein, 2014). Quantitative approaches such as cost–benefit analysis and integrated models improve comparability and aggregation, but they similarly rely on structured and complete datasets (Fadiga and Katjiuongua, 2014; Mishra, 2018; Welde and Volden, 2018; de Jong, Vignetti and Pancotti, 2019; Ponanan and Satirasetthavee, 2025). Qualitative approaches instead focus on context, processes, and mechanisms of change using interviews, documents, and local knowledge, making them more adaptable to incomplete data environments but less consistent in terms of comparability (Späth, 2008; Crespo, 2023; Dörrbecker, 2023).

Theory-based frameworks like Contribution Analysis and Process Tracing offer promising ways to reconstruct causal explanations by combining multiple data sources, but they share a common vulnerability. Whether relying on statistical identification, structured comparison, or causal reconstruction, they all assume a minimum level of coherence in the available evidence (Rogers and Weiss, 2007; Mayne, 2012). This assumption is often not met. Some evaluations suffer from missing baselines, weak documentation, or limited institutional continuity, while others face the opposite problem of excessive but fragmented and poorly integrated information. In both cases, the main challenge is not the absence of methods, but the difficulty of working with evidence that

is dispersed, uneven, and not naturally structured for program-level analysis (Nicolaisen and Driscoll, 2014; Welde and Volden, 2018; Dörrbecker, 2023). Even mixed-method approaches, while improving triangulation, do not fully resolve the challenge of integrating heterogeneous evidence into a single coherent evaluative structure (Bhagwat, Pathak and Venkataramani, 2011; Knook *et al.*, 2018; Schimpf, Barbrook-Johnson and Castellani, 2021).

To conclude this section, our research suggests that the core methodological gap is not the lack of evaluation approaches, but the absence of a systematic mechanism for integrating incomplete, fragmented, and heterogeneous project-level evidence into a coherent ex-post program-level evaluation. This challenge persists across methodological traditions and under varying information conditions, and remains insufficiently addressed in the current evaluation landscape.

4. SYNTHESIS AND RESEARCH PROSPECTS: TOWARD A NEW EVALUATIVE FRAMEWORK

4.1 Methodological Implications Emerging from the Literature

Ex-post program evaluation requires frameworks capable of operating under imperfect and variable information conditions. Studies reveal that the main challenge is less the choice of a method than the production of coherent evaluative conclusions based on data that is often incomplete and fragmented across projects and organizational systems (Langbein, 2014).

A first implication is the need for flexibility in evaluation design. Ex-post evaluations are frequently constrained by time, budget, institutional continuity, and data availability (Bamberger *et al.*, 2004; Späth, 2008; Longhurst, 2013). In such conditions, highly demanding procedures are difficult to apply consistently. These evaluations need to remain adaptable to different resource and information environments while clearly stating their assumptions and limitations (Bamberger *et al.*, 2004; Longhurst, 2013).

A second implication concerns the combination of multiple forms of evidence to improve the interpretation of impacts and program success. Mixed-method approaches are widely considered useful because they integrate quantitative measurement with contextual and explanatory understanding, thereby improving coverage and generalizability (Garbarino and Holland, 2009; Castaño, Blanco and Martinez, 2019). Participatory approaches and triangulation are repeatedly put forward to strengthen credibility when information is incomplete or uneven (Bamberger *et al.*, 2004; Garbarino and Holland, 2009; Longhurst, 2013). Within this logic, recall-based techniques can also

be valuable when combined with other sources of information and systematically triangulated (Bamberger *et al.*, 2004). Regardless of the methodological approach chosen, many authors underline that it should be applied consistently to the full set of available data. This systematic use across multiple similar projects enhances learning and comparability (Welde and Volden, 2018; Samset and Volden, 2022).

A third implication is related to the role of evaluator judgement and uncertainty in ex-post assessment. Evaluation findings are shaped not only by available data, but also by interpretation, selection of criteria, and evaluative reasoning processes (Langbein, 2014; Dörrbecker, 2023). Even standardized frameworks and scoring systems cannot fully eliminate subjectivity, although they may improve consistency by structuring interpretation and reducing ambiguity in scoring procedures (Casper *et al.*, 2020; Dörrbecker, 2023). Accordingly, several authors stress the importance of transparency, reflexivity, and explicit strategies for acknowledging and addressing uncertainty and potential bias throughout the evaluation process (Crespo, 2023; Dörrbecker, 2023).

Finally, ex-post evaluation frameworks should account for the multidimensional nature of development interventions. Programs are rarely assessed according to a single criterion alone, as effectiveness may involve economic, social, organizational, environmental, and institutional dimensions simultaneously (Langbein, 2014; Welde and Volden, 2018). This calls for approaches capable of integrating multiple evaluative perspectives while remaining sufficiently coherent to support comparison, synthesis, and organizational learning across projects and contexts (Samset and Volden, 2022).

In sum, these elements suggest that no single evaluation approach is likely to be universally applicable across all contexts (Hallam and Bonino, 2013; Castaño, Blanco

and Martinez, 2019). Instead, the evidence points toward adaptable frameworks that balance structure and flexibility, comparability and contextual interpretation, as well as methodological rigor and operational feasibility under real-world evaluation conditions.

4.2 Program Reconstruction Under Fragmented Evidence in Data Limited Conditions

Reconstructing a program after implementation in an international development setting involves rebuilding a coherent account from information that is incomplete and scattered across multiple sources. The challenge is not simply a lack of data, but rather that information is dispersed across reports, formats, and time periods, and was not originally designed to be combined at program level (Nicolaisen and Driscoll, 2014; Welde and Volden, 2018; Dörrbecker, 2023). Under these conditions, simply bringing together available evidence is often insufficient to understand what the program actually consisted of, how it evolved, and what it ultimately achieved.

4.2.1 Misalignment of Evaluation Frameworks Across Project and Program Levels

International development programs are commonly represented through theories of change and logic models, largely because these tools are required by donors at the design stage (Rogers and Weiss, 2007; Hermano *et al.*, 2013; Mishra, 2018). A theory of change is intended to explain how and why change is expected to occur, while logic models translate this reasoning into a more operational structure linking inputs, activities, outputs, outcomes, and impacts. While useful for planning and reporting, these models often simplify reality into linear sequences that do not clearly capture the mechanisms,

assumptions, or alternative pathways through which change actually occurs (Rogers and Weiss, 2007; Hermano *et al.*, 2013). This can lead to an incomplete or sometimes misleading picture of how programs function in practice.

These limitations also shape how evaluation frameworks such as the OECD-DAC criteria (relevance, coherence, efficiency, effectiveness, impact, and sustainability) are applied at both project and program levels (Glewwe and Todd, 2022; Chasanah, Gunawan and Baroudi, 2024). In principle, the same evaluation criteria can be applied across both levels (OECD, 2025), but this creates ambiguity when the intervention structure itself is not clearly defined. In real evaluation settings, these criteria are often applied to interventions whose underlying logic models remain too broad or only partially operationalized. This widens the gap between the formal structure of evaluation frameworks and the complexity of real implementation processes. At times, there is no consistent way to connect multiple projects into a single program-level view. In other cases, the framework is used in a simplified manner, where criteria function more as labels than as analytical tools (Samset and Volden, 2022).

In this context, evaluations may look methodologically rigorous while still offering limited insight into what actually happened. Efficiency, for instance, is often narrowed to cost considerations, while impact is discussed without sufficient contextual or causal grounding (Samset and Volden, 2022). Indicators typically capture outputs and, at best, some outcomes, but they rarely allow robust conclusions about broader or long-term effects, especially when they are not yet observable at the time of measurement (Dörrbecker, 2023). The literature also shows that even when formal frameworks exist, their underlying logic often contains missing causal links or unrealistic assumptions in

goal hierarchies (Samset and Volden, 2022). Together, these limitations illustrate how evaluation frameworks can remain formally coherent while still providing only a partial understanding of program reality.

4.2.2 Project Heterogeneity and its Implications for Program Reconstruction

To reach a sufficient level of detail without being overwhelmed by fragmented low-data evidence, retrospective program reconstruction needs to connect three levels: deliverables, projects, and the overall program. Programs are mainly composed of multiple projects, while projects themselves consist of concrete deliverables produced within a defined timeframe (Schiuma, 2015; Levin and Ward, 2016; Palomares, 2019; Dijkzeul and Salomons, 2021). Reconstruction therefore starts with identifying and classifying deliverables, then linking them back to their originating projects, and finally organizing these projects into a coherent program-level structure. In this process, a systems perspective can help coordinate the overall structure and improve coherence across multiple components (Eigbe, Sauser and Felder, 2015).

Projects, however, differ widely. They can be classified according to the clarity of their objectives and methods, ranging from well-defined engineering projects to more uncertain product development, software, or organizational change initiatives (Shenhar *et al.*, 2001; Khang and Moe, 2008; Ahsan and Gunawan, 2010; Eigbe, Sauser and Felder, 2015). These differences imply distinct management and evaluation approaches, yet no single classification system has reached broad consensus, reflecting the diversity and complexity of real-world projects (Stretton, 2014a, 2014b, 2014c, 2014d, 2017, 2021).

This means project reconstruction cannot rely on a fixed typology. Instead, the set of projects within a program must first be examined empirically and then interpreted using existing classifications as flexible analytical references rather than rigid categories.

This distinction matters because evaluation dimensions do not apply uniformly across all project types. Projects vary in size, uncertainty, stakeholder diversity, interdependencies, institutional context, and expected results (Youker, 2003; de Souza Pinto *et al.*, 2014; Azmat and Siddiqui, 2023). They may also reflect different definitions of success, including time, cost, quality, stakeholder satisfaction, long-term benefits, or organizational learning (Shenhar *et al.*, 2001; Diallo and Thuillier, 2004; Chasanah, Gunawan and Baroudi, 2024). Since no single definition of project success is universally accepted (Chasanah, Gunawan and Baroudi, 2024), reconstruction must first clarify what types of projects exist within the program before determining which evaluation dimensions are relevant to each category. The purpose is therefore not only to regroup fragmented evidence, but also to build a coherent analytical structure that supports subsequent program-level evaluation. This makes program reconstruction a necessary step to restore analytical coherence across fragmented evidence and enable meaningful application of evaluation frameworks at program level. Then, this reconstructed program structure provides the analytical foundation upon which standardized evaluative dimensions and anchored ordinal assessment grids are subsequently applied.

Table 2. Methodological Implications from Section 4.1 and Program Reconstruction Responses in Section 4.2

Methodological Implications from Section 4.1	Program reconstruction responses proposed in Section 4.2
Development interventions involve multiple evaluative dimensions simultaneously (economic, social, organizational, environmental, and institutional), making single-dimensional assessment insufficient	Reconstruction re-establishes a structured analytical unit by linking deliverables → projects → program, creating the basis for connecting project-level dimensions with OECD program evaluation criteria and supporting multidimensional interpretation
Evaluation approaches must remain sufficiently coherent to support comparison, synthesis, and organizational learning across projects and contexts)	Reconstruction explicitly addresses project heterogeneity by identifying project types and structuring them within a common program-level analytical framework, allowing consistent application of evaluation dimensions derived from both literature and empirical evidence

4.3 Structured Evaluation Interpretation Through Anchored Dimensions

Ex-post program evaluation often requires judgement to be built from evidence that is only partially available and unevenly distributed across projects and organizational sources (Nicolaisen and Driscoll, 2014; Welde and Volden, 2018; Dörrbecker, 2023). In such contexts, evaluation cannot rely exclusively on standardized quantitative procedures, but neither can it depend entirely on open-ended qualitative interpretation.

The framework proposed here relies on anchored evaluative dimensions. Each project dimension is assessed through an ordinal grid made of clearly differentiated qualitative anchors designed to limit overlap between categories (Casper *et al.*, 2020). Ordinal scales are therefore used primarily as structured interpretive tools rather than statistical measurement instruments (Agresti, 2010; Martinez-Martin, 2010). The objective is not to transform qualitative judgement into artificial numerical precision, but to improve consistency and comparability across evaluations (Langbein, 2014; Crespo, 2023). This view is consistent with research on ordinal measurement showing that overlapping verbal anchors can increase ambiguity and produce inconsistent

interpretation, especially when categories vary across evaluators and contexts (Casper *et al.*, 2020).

Because evidence quality varies considerably across projects and sources, each ordinal assessment is paired with a level of evidential certainty. Rather than treating all evidence as equally reliable, certainty assessments reflect the robustness of the information supporting each evaluative judgement (Lepskiy, 2023). This remains consistent with work arguing that evaluative structures must stay operational and explicit under real-world resource and information constraints (Bamberger *et al.*, 2004; Longhurst, 2013).

Table 3. Methodological Implications from Section 4.1 and Anchored Evaluative Responses in Section 4.3

Methodological Implications from Section 4.1	Anchored Evaluative Responses in Section 4.3
Evaluation operates under constraints of time, budget, institutional continuity, and data availability, requiring flexible but transparent evaluation designs	Anchored dimensions provide a stable but adaptable structure applicable under varying resource and information conditions while keeping assumptions explicit
Reliable evaluation requires combining multiple forms of evidence, including interviews, documentary analysis, and triangulation of heterogeneous sources	Project-level scores are derived through structured triangulation of qualitative and documentary evidence (interviews, reports, and archival sources), enabling systematic integration of heterogeneous information into ordinal anchors
Evaluation outcomes are influenced by evaluator judgement, requiring mechanisms to reduce ambiguity and improve consistency without eliminating subjectivity	Standardized ordinal anchors and explicitly defined criteria reduce interpretive variability while maintaining reflexive judgement and improving cross-project comparability

4.4 Relational Data Integration and Analytical Perspectives

Extending the methodological foundations developed in the previous sections, this research proposes a relational data architecture for ex-post program evaluation under fragmented, heterogeneous, and incomplete information conditions. The aim is not only to store evaluation data, but to organize dispersed project-level evidence in a way that preserves traceability and consistency at program level (Azuddin *et al.*, 2025). Database design research shows that complex information systems require clearly defined entities, relationships, referential integrity, and normalization to remain coherent and analytically usable (Harrington, 2016; Meier and Kaufmann, 2019). Relational databases are particularly suited to this purpose because they organize data into interconnected tables rather than isolated records, enabling links across levels of analysis (Azuddin *et al.*, 2025).

Within this structure, information is organized around entities reflecting development program structures, such as projects, program components, deliverables, partners, and volunteers. This makes it possible to situate each intervention within its organizational and functional context. The architecture is designed for environments where information remains fragmented and weakly integrated across systems. As noted by Kitchin and Lauriault (2015) and Hekler *et al.* (2019), the main challenge is often not data volume, but the difficulty of connecting dispersed information into a coherent analytical structure.

A central feature of the model is the explicit representation of uncertainty in project evaluation scores. As discussed in section 4.3, each evaluative entry is associated with a certainty score. This follows the view that uncertainty is inherent to real-world

data rather than an anomaly to eliminate (Nunes and Sousa, 2009), and that it should be represented explicitly in relational systems handling incomplete information (Link and Prade, 2016). Uncertainty is therefore attached to evaluative judgements rather than being handled at the level of probabilistic modelling within the database itself (Suciu *et al.*, 2022).

Once organized, the dataset supports a second analytical layer combining ordinal analysis and explainable machine learning. Ordinal data are treated as ranked information rather than continuous measurements, avoiding assumptions that are not warranted by the nature of the data (Casper *et al.*, 2020; Schnuerch *et al.*, 2022; Brauer and Day, 2025). This direction aligns with recent applied work that integrates expert-based ordinal scoring systems with machine learning approaches. Bhattacharya and Florez-Arango (2025), for example, use the Mobile App Rating Scale (MARS) as an ordinal expert evaluation framework combined with clustering and supervised learning to analyze how evaluative dimensions relate to observed outcomes. Rather than focusing on prediction, their work highlights the relative explanatory value of evaluation dimensions across contexts.

In this study, the objective is similarly not prediction, but the identification of interpretable relationships between project characteristics, contextual factors, and evaluative outcomes. This is particularly relevant in fragmented data environments where standard statistical assumptions are often only partially met. Explainable machine learning is therefore used to support interpretation. The emphasis is placed on identifying variables associated with variation in project scores and examining how these relationships differ across intervention types and program components (Belle and

Papantonis, 2021). This helps maintain transparency and coherence with the overall evaluative logic of the framework.

Table 4. Methodological Implications from Section 4.1 and Relational Architecture and Analytics Responses in Section 4.4

Methodological Implications from Section 4.1	Relational Architecture and Analytics Responses in Section 4.4
Need for flexibility in evaluation design under resource, institutional, and data constraints	A relational structure that can accommodate incomplete and heterogeneous program information without requiring full data availability
Need for combining multiple forms of evidence while maintaining coherence	Integration of heterogeneous data sources within a unified relational schema to ensure consistency across levels of analysis
Importance of evaluator judgement and transparency in interpretation	Structured outputs that support interpretation while keeping human judgement central rather than fully automated decision-making
Need for multidimensional evaluation across economic, social, institutional, and organizational dimensions	Representation of multiple program dimensions through decomposition into relational entities and components
Requirement for balancing methodological rigor and operational feasibility	A simplified relational and analytical design that prioritizes interpretability and usability over complex or fully data-intensive approaches

CONCLUSION

In summary, this literature review shows that ex-post program evaluation in international development has developed into a wide range of approaches, but they continue to face the same underlying challenge: evidence is often fragmented, incomplete, and dispersed across multiple projects and sources. This makes it difficult to reconstruct a coherent program-level understanding of what has been achieved and how different components contribute to overall results. This gap between theoretical expectations and the structure of available evidence remains a central issue in the field.

Methodologically, the review highlights a clear tension between formal evaluation approaches and the complexity of implementation contexts. Some approaches prioritize standardization, comparability, and clear measurement rules, which require relatively complete and well-organized data. Others focus more on interpretation, context, and flexibility, allowing them to work with more uneven evidence but limiting their ability to systematically compare and synthesize results across cases. Overall, this leads to a situation where evaluation practice must balance rigor with feasibility, without a fully satisfactory way of combining both across different types of projects and information conditions.

To address this gap, an integrated approach is needed to structure, connect, and make sense of this dispersed evidence. The theoretical framework developed in this review provides the basis for the research that follows. Building on this synthesis, the next steps will focus on turning these ideas into a practical method, which will be tested and refined using empirical data to improve clarity and consistency in ex-post evaluation within development contexts.

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